

Program : Diploma in Tool & Die Engineering	
Course Code : 4111	Course Title: Fluid Mechanics & Heat Transfer
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To establish fundamental knowledge of basic fluid mechanics and address specific topics relevant to simple applications involving fluids.
- To familiarize students with the relevance of fluid dynamics to many engineering systems.
- To appreciate basics of hydraulic and pneumatic systems
- To introduce the basics of thermodynamics and modes of heat transfer

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Physics		Applied Physics I & II	1 & 2
Engineering Science		Engineering Mechanics	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate fluid properties, pressure measuring devices and hydrostatic forces	15	Applying
CO2	Investigate the applications of laws of conservation of momentum, mass and energy in fluid flow systems and the characteristics of flow through pipes, orifices and notches	14	Applying
CO3	Distinguish the components and basic circuits in hydraulic and pneumatic systems	15	Understanding
CO4	Illustrate the basic thermodynamics principles and different modes of heat transfer	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	3					
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate fluid properties, pressure measuring devices and hydrostatic forces		
M1.01	Explain fluid properties	3	Understanding
M1.02	Explain the Fluid Pressure and the methods of measurement	4	Understanding
M1.03	Illustrate the principle of working of various pressure measuring devices and solve simple problems on those	4	Applying
M1.04	Interpret the terms total pressure & buoyancy and calculate the total pressure on immersed surfaces	4	Applying
Contents: Properties of fluid: Density, Specific gravity, Specific Weight, Specific Volume, Dynamic Viscosity, Kinematic Viscosity, Surface tension, Capillarity, Vapour Pressure, Compressibility. Numerical problems on fluid properties. Fluid Pressure & Pressure Measurement: Fluid pressure, Pressure head, Pressure intensity, Concept of vacuum and gauge pressures, atmospheric pressure, absolute pressure, Simple and differential manometers, Bourdons pressure gauge, Concept of Total pressure on immersed bodies, center of pressure, Simple problems on pressure measurements Buoyancy and flotation, Archimedes' principle, stability of immersed and floating bodies, determination of metacentric height. Problems on total pressure.			
CO2	Investigate the applications of laws of conservation of momentum, mass and energy in fluid flow systems and the characteristics of flow through pipes, orifices and notches		
M2.01	Describe the types of fluid flows	2	Understanding

M2.02	Apply the momentum and energy equations to fluid flow problems based on an analysis of the various system specifications	4	Applying
M2.03	Illustrate the concept of flow through pipes, orifices & notches	4	Applying
M2.04	List the losses of head in pipes and identify Major losses and Minor losses and solve problems	4	Applying
	Series Test I	1	

Contents:

Fluid Flow: Types of fluid flows, Path line and Stream line, Continuity equation, Bernoulli's theorem, Principle of operation of Venturimeter, Orifice meter and Pitot tube, Derivations for discharge, coefficient of discharge, numerical problems.

Flow Through Pipes, Orifices & Notches: Darcy's equation and Chezy's equation for frictional losses (No derivation), Minor losses in pipes, Numerical problems to estimate major and minor losses. Orifices: types of orifices, Vena contracta, coefficient of contraction, coefficient of velocity, coefficient of discharge, simple problems. Notches: Types –Rectangular notches , triangular notch , trapezoidal notch , discharge over notches – equations only, simple numerical problems

CO3	Distinguish the components and basic circuits in hydraulic and pneumatic systems		
M3.01	Discuss basics of Hydraulics and Pneumatics and its applications	1	Understanding
M3.02	Summarize the features and functions of various components of hydraulic and pneumatic systems	4	Understanding
M3.03	Identify various Hydraulic and Pneumatic symbols	2	Remembering
M3.04	Describe the types of valves used in hydraulic and pneumatic systems	4	Understanding
M3.05	Explain the working of different types of hydraulic and pneumatic circuits and systems	4	Understanding

Contents:

Hydraulics: Introduction, basic concepts, Hydraulic fluids, Classification and properties of hydraulic fluids, Basic components, Fluid power generators, i.e. Gear, Vane, Piston pumps, linear and Rotary Actuators, Direction Control Valves, types, actuation methods, pressure control valves; pressure reducing valves, pressure relief valve, Flow control valves - simple and pressure compensated type. Circuit symbols, automatic cylinder reciprocating circuit & hydraulic circuit for robotic arm.

Pneumatics: Introduction, Basic components, Source, storage and distribution, treatment of compressed air, linear and Rotary actuators, Direction control valves – types, actuation methods, pressure control valves, logic devices – twin pressure valve, shutter valve, time delay valve, Basic pneumatic circuits, principle of working of power operated holding devices – chuck and clamping device, Pneumatic circuit design and analysis			
CO4	Illustrate the basic thermodynamics principles and different modes of heat transfer		
M4.01	Explain basic concepts in thermodynamics	3	Understanding
M4.02	Interpret the concepts of conduction to solve problems involving steady state heat conduction in plane walls.	6	Applying
M4.03	Illustrate concepts of convection and radiation	5	Applying
	Series Test II	1	
Contents: Introduction to Thermodynamics : Role of Thermodynamics in Engineering and Science, Types of Systems, Thermodynamic Equilibrium, Properties, State, Process and Cycle, Elementary introduction to Zeroth, First and Second laws of thermodynamics, Heat and Work Interactions for various non-flow and flow processes Heat Transfer: Conduction, convection and radiation-Fourier's law of thermal conduction. Thermal conductivity, conduction through a plane wall and composite plane wall – Numerical problems. Thermal radiation – reflection, absorption and transmission, absorptivity, reflectivity and transmissivity, Stefan - Boltzman's law of total radiation. Newton Rikhman equation of Thermal convection, free convection, forced convection. Simple numerical problems on convection and radiation.			

Text/ Reference:

T/R	Book Title/Author
T1	Rattan S S, “Fluid Mechanics & Hydraulic Machines”, Khanna Publishing House, New Delhi, 2019
T2	Yunus. A Cengel and Afshin. J Ghajar, “Heat and Mass Transfer: Fundamentals and Applications”, McGraw-Hill Education
T3	Nag P K, “Engineering Thermodynamics”, McGraw-Hill Education, Sixth Edition 2017
R1	Ramamritham. S, “Hydraulics, Fluid Mechanics And Fluid Machines”, DhanpatRai&Sons, Delhi, 2006.
R2	P. N Modi and S. M. Seth, “Hydraulics and Fluid Mechanics Including Hydraulics Machines”, 22nd Edition, Standard Book House, 2019
R3	Anthony Esposito, “Fluid Power with applications”, Pearson Education.
R4	Werner Deppert and Kurt Stoll, “Pneumatic Control-An introduction to the principles”, Vogel-Verlag.
R5	Rajput R K, “A Textbook of Heat and Mass Transfer” S. Chand Publications, 2019

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/112104118/
2	https://freevideolectures.com/course/89/fluid-mechanics
3	http://www.efunda.com/formulae/heat_transfer/home/overview.cfm
4	http://www.nptelvideos.in/2012/12/heat-and-mass-transfer.html
5	http://web.mit.edu/hml/ncfmf.html